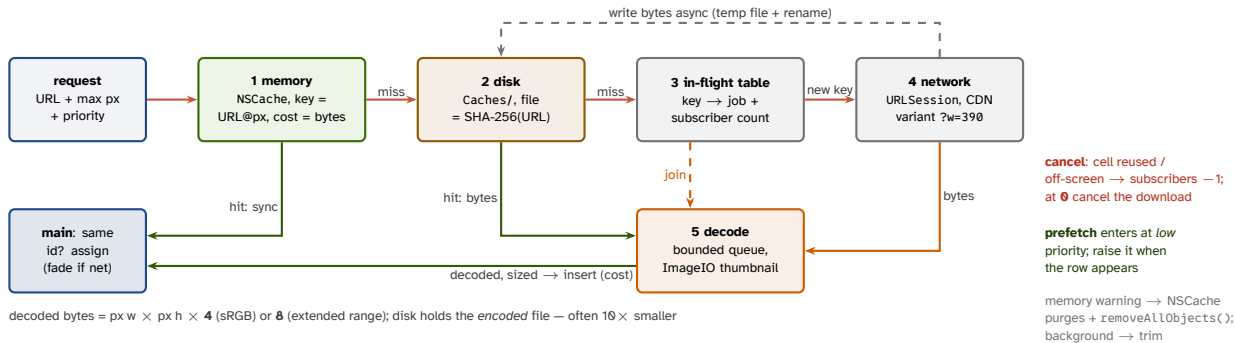


**In one line:** A request walks **memory** (decoded, *sized* bitmaps) → **disk** (encoded bytes) → **network** (deduplicated, prioritised, cancellable) and comes back through **decode + downsample off the main thread**. What matters: the cache **key**, the **memory maths**, **one download per key** with cancellation only when the *last* subscriber leaves, an **identity check** before assigning, and **prefetch** at low priority.



## How it works

- Key** = URL + target *pixel* size (+ transforms: crop, rounding, blur). Same URL at 2 sizes = 2 memory entries; disk keeps one set of bytes per URL.
- Memory:** NSCache<Key, UIImage> — thread-safe, keys not copied, evicts on its own under pressure. Set `totalCostLimit` with cost = decoded bytes; `countLimit` alone lets 10 huge images blow the budget. Apple: eviction order “not guaranteed”, limit “not strict” — so not an LRU; size it, don’t rely on order.
- Disk:** encoded bytes in Caches/ (purgeable, not backed up); index of size + last-access; sweep to a byte cap + max age, off main (launch, background); write atomically. **URLCache vs custom:** URLCache gives HTTP semantics (Cache-Control, ETag revalidation) but only stores 2xx responses whose headers allow it and that are ≤ ~5% of its disk capacity, and you do not choose eviction. Custom: your key, LRU, size cap, pinning of prefetched items. Neither holds a *decoded* tier.
- Network:** **coalesce** — N cells asking for one key share one job; each caller is a subscriber; cancel the transfer only when the count reaches 0. **Priority:** URLSessionTask.priority (0-1; lowPriority 0.25, highPriority 0.75; only a *hint*) — visible high, prefetch low. Cap concurrency. Ask the CDN for the size you need (?w=390, Accept: image/webp) — the cheapest byte is the one never sent.
- Decode:** off main on a *bounded* queue (WWDC18: a serial queue, not one global-queue block per image → thread explosion). ImageIO CGImageSourceCreateThumbnailAtIndex + kCGImageSourceThumbnailMaxPixelSize, or iOS 15 byPreparingThumbnail(ofSize:). A plain UIImage(data:) decodes *later*, full size, on main, in the commit’s prepare phase.
- Cancellation:** prepareForReuse / didEndDisplaying; SwiftUI .task(id: url) cancels on disappear or id change. Then **check identity** before assigning — a late result must not land in a reused cell.
- Prefetch:** UICollectionViewDataSourcePrefetching (iOS 10) prefetchItemsAt / cancelPrefetchingForItemsAt; it is a *hint* — not called for every row on a fast fling, so display must work without it. SwiftUI lazy stacks have no prefetch hook: warm row  $n+k$  from row  $n$ ’s onAppear.
- Placeholders:** dominant colour or **BlurHash** (a ~20-30 char string in the JSON, decoded to a tiny image) → progressive JPEG (CGImageSourceCreateIncremental) → final. Fade in only on a network load, never on a memory hit (flicker on every scroll).
- Formats:** HEIC iOS 11 · WebP decode iOS 14 · AVIF iOS 16 (*unverified*) · JPEG XL iOS 17. Check at runtime: CGImageSourceCopyTypeIdentifiers(). Animated GIF/WebP = one bitmap per frame in memory.

## Example — the cell side

```
final class PhotoCell: UICollectionViewCell { // @MainActor
    private var load: Task<Void, Never>; private var shownID: Photo.ID?
    func show(_ p: Photo, _ pipe: ImagePipeline) {
        shownID = p.id; imageView.image = p.blurHash // instant
        let px = max(bounds.width, bounds.height)
            * traitCollection.displayScale // PIXELS
        load = Task { // inherits main actor
            let img = try? await pipe.image(p.url, maxPixel: px, .high)
            guard !Task.isCancelled, shownID == p.id, let img else { return }
            imageView.image = img // right cell, main
        }
        override func prepareForReuse() {
            super.prepareForReuse(); load?.cancel(); imageView.image = nil
        }
    }
}
```

pipe.image must translate the caller’s cancellation into “subscriber —1” — cancelling the shared task would kill the other cells’ load.

## Memory maths — say the numbers

| image                           | decoded                      | × 60 cells      |
|---------------------------------|------------------------------|-----------------|
| 12 MP photo, 4032 × 3024        | 4032·3024·4 = <b>48.6 MB</b> | 2.9 GB — jetsam |
| 3-col grid, 130 pt @3x = 390 px | 390·390·4 = <b>608 KB</b>    | 36 MB           |
| same, extended range (8 B/px)   | 1.2 MB                       | 73 MB           |

## Traps

- ! Memory cache of *encoded* data or full-size UIImage(data:) — decode on main, every first display, at full resolution.
- ! Cancelling the download when *one* of three cells showing it is reused.
- ! Prefetch at default priority — it competes with what is on screen.
- ! 50 decodes dispatched at once on global queues — thread explosion, worse latency.
- ! “Just use URLCache” — no decoded tier, the ~5% size rule, eviction you do not control.
- ! Fade-in animation on a memory hit; no identity check → wrong image flashes.

## Remember

**Key · tier · dedupe · decode small off main · check identity · prefetch low.** Memory = pixels × 4; disk = bytes; network = the size you need.

*Related:* ios-client-system-design (feed) · rendering-pipeline (downsample code, lazy decode) · instruments-performance (memory) · collectionview-compositional · data-transfer-optimisation (formats)